Python While Loop – 10 Examples

What is a while loop?

A while loop is used to repeatedly execute a block of code as long as a given condition is True.

Syntax:

```
while condition:
    # code to execute
```

1. Print numbers from 1 to 10

```
i = 1

while i <= 10:
    print(i)
    i = i + 1
```

2. Print even numbers from 1 to 20

```
i = 1

while i <= 20:
    if i % 2 == 0:
        print(i)
    i = i + 1
```

3. Print odd numbers from 1 to 20

```
i = 1

while i <= 20:
    if i % 2 != 0:
        print(i)
    i = i + 1
```

4. Sum of numbers from 1 to 10

```
i = 1
total = 0

while i <= 10:
    total = total + i
    i = i + 1

print("Sum =", total)
```

5. Multiplication table

```
num = 5
i = 1

while i <= 10:
    print(num, "x", i, "=", num * i)
    i = i + 1
```

6. Count digits in a number

```
num = 12345
count = 0

while num > 0:
    count = count + 1
    num = num // 10

print("Digits =", count)
```

7. Reverse a number

```
num = 12345
reverse = 0

while num > 0:
    digit = num % 10
    reverse = reverse * 10 + digit
    num = num // 10

print("Reverse =", reverse)
```

8. Check palindrome number

```
num = 121
original = num
reverse = 0

while num > 0:
    digit = num % 10
    reverse = reverse * 10 + digit
    num = num // 10

if original == reverse:
    print("Palindrome")
else:
    print("Not Palindrome")
```

9. Factorial of a number

```
num = 5
factorial = 1
i = 1

while i <= num:
    factorial = factorial * i
    i = i + 1

print("Factorial =", factorial)
```

10. Check prime number

```
num = 7
i = 2
prime = True
```

```
while i < num:
    if num % i == 0:
        prime = False
        break
    i = i + 1

if prime and num > 1:
    print("Prime number")
else:
    print("Not a prime number")
```

Important Points

• The while loop continues while its condition is True.

• Update the loop variable to avoid an infinite loop.

• break is used to stop the loop immediately.

• while loops are useful when the number of repetitions is not fixed.